



Saathi

A Companion app for Senior Citizens



Milestone - 3

Software Engineering Project

SE [CS3001]

Team-19

Table of Contents

1. Sprint Schedule	3
2. SCRUM: Minutes of Meeting.....	4
3. Scheduling and Project Planning.....	6
Kanban Board(Partial)	
Gantt Chart	
4. Class Diagram.....	8
5. Components' Design.....	9
6. Frontend Designs.....	10

Team Profile

S.No	Name	Student Email
1.	Devaanshi Chaudhary	21f3001136@ds.study.iitm.ac.in
2.	Advait Vats	21f1005152@ds.study.iitm.ac.in
3.	Somsubhra De	22f3002680@ds.study.iitm.ac.in
4.	Drishti	21f1004302@ds.study.iitm.ac.in
5.	Shourjya Bhattacharjee	21f1000757@ds.study.iitm.ac.in
6.	Karthik Ganesh	21f2000775@ds.study.iitm.ac.in
7.	Jatin Yadav	22f2001018@ds.study.iitm.ac.in

1. Sprint Schedule

Sprint #1: Team Formation and Topic Selection

Team registration, finalising the topic, identifying potential users.

2025, May.12th - May.20th

Sprint #2: User Stories

Interviewing potential users & writing user stories; finalising documentation for Milestone 1 report.

2025, May.20th - May.24th

Sprint #3: Creating Storyboards

Creating storyboards using the user stories, creating low-fidelity wireframes and finalising Milestone 2 report.

2025, May.26th - June.6th

Sprint #4: Scheduling and Design Components

Develop project timeline, document the tools used for scheduling, start working on frontend pages; finalising Milestone 3 report.

2025, June.20th - July.4th

Sprint #5: Implement API and Integrating GenAI

Creation of API and documenting them in a YAML file, for API endpoints.

2025, July.7th - July.14th

Sprint #6: Integrating APIs

Integrating api in the app with the app. YAML file for API endpoints, finalise Milestone 4 report.

2025, July.14th - July.17th

Sprint #7: Designing Test Cases

Designing test cases, testing cases for API and other functionalities.

2025, July.20th - July.31st

Sprint #8: Creating Basic Unit testing

Create basic unit test cases; create, review and finalise Milestone 5 report.

2025, July.31st - August.4th

Sprint #9: Finalising Projects

Recording demo for the final project with documenting all the tools and technology to prepare for the project presentation.

2025, August.5th - August.15th

2. SCRUM: Minutes of Meeting

Mode of meetings: *Google Meet*

Date and time: Meetings were held twice a week at *7pm- 8pm*.

Attendees:

- Devaanshi Chaudhary
 - Advait Vats
 - Somsubhra De
 - Drishti
 - Shourjya Bhattacharjee
 - Karthik Ganesh
 - Jatin Yadav
-

Sprint #1: Team Formation and Topic Selection

Wednesday, 14 May 2025

The team convened to discuss and finalise the project topic. After evaluating various ideas based on feasibility, relevance, and technical scope, a consensus was reached on a suitable project direction and selecting question 1 as project topic. Following this, the team filled and submitted the official team registration form. Roles were assigned among team members, including designation of the scrum master, project architect and development responsibilities(frontend, backend and testing).

Sprint #2: User Stories and Creating Storyboards

Saturday, 17 May 2025

During the Saturday session, the team conducted interviews with four potential users from the target demographic. These interviews provided valuable insights into user needs, expectations, and common challenges, which were documented for further reference. In the following meeting, held on Wednesday, the team collaboratively drafted user stories based on the feedback. These stories went through a peer review process, were refined for clarity and relevance, and finalized for inclusion in project documentation.

Sprint #2.1: Milestone 1

Wednesday, 21 May 2025

The focus of this sprint was to consolidate all Milestone 1 activities. On Wednesday, the team reviewed the finalised user stories and interview notes. The milestone report draft was prepared and internal feedback was incorporated. On Saturday, the final milestone document was submitted successfully. A short retrospective was conducted where the team reflected on the planning and execution of initial sprints, identifying strengths such as collaboration and timely communication, while also noting the need for improved time management.

Sprint #3.1: Wireframes and Storyboard

Saturday, 24 May 2025

The team transitioned into the design and technical planning phase. During the 28 May meeting, low-fidelity wireframes for key screens—such as Home, Dashboard, and Login—were created using streamlit (frontend) and fast api (backend). A basic storyboard was also sketched to visualise the screen flow and user navigation paths. On 31 May, initial setup of the application structure was carried out using Streamlit, and a first draft of the database schema was created, including an ER

diagram. In the 4 June meeting, feedback was discussed, leading to adjustments in both the storyboard and schema to better align with functional requirements.

Sprint #3.2: Milestone 2

Saturday, 7 June 2025

Finalisation tasks for Milestone 2 began with a team review of all prototypes, including updated wireframes, storyboards, and the database schema. The 7 June session was focused on compiling and formatting the milestone documentation. On 11 June, the team held a quality check session to address formatting inconsistencies and minor design flaws. The final submission was completed during the 14 June meeting. A sprint retrospective followed where the team noted improved coordination in design planning, and agreed to maintain this structure in future sprints.

Sprint #4.1: UI elements, Pages, Class Diagram

Wednesday, 18 June 2025, Saturday, 21 June 2025, Wednesday, 25 June 2025, Saturday, 28 June 2025, and Wednesday, 2 July 2025

This milestone involved active UI development and early functional integration. On 18 June, frontend design elements such as buttons, navigation bars, and sidebars were built using CSS and Streamlit components. The structure of the frontend was also defined at this point. During the 21 June session, the team began coding core pages like the dashboard and homepage. Simultaneously, the class diagrams for backend integration were sketched. The 25 June meeting focused on implementing and testing a basic login and sign-up system, including mock session management. On 28 June, database connectivity was finalized, and internal testing of all UI components was initiated. The session on 2 July addressed bug fixing, with a focus on UI responsiveness and login validation. All components were polished and aligned to match the initial wireframes.

Sprint #4.2: Milestone 3

Wednesday, 2 July 2025 and Saturday, 5 July 2025

The final milestone submission activities were carried out in this sprint. On 2 July, a full project walkthrough was conducted. Peer QA reviews were done to identify issues across components, and all required fixes were implemented. The Saturday meeting on 5 July was utilized to complete the final submission of all documents and code. A detailed retrospective followed, summarizing sprint outcomes, acknowledging effective practices, and documenting areas for improvement in future iterations or real-world deployment.

3. Scheduling and Project Planning

We selected **Jira** as our central project management tool because of its customisable workflow schemes, and integrated sprint planning features which allowed us to keep every backlog item and requested feature neatly organised and easily accessible.

Within Jira, the Kanban board were used to visualise for tracking the status of individual tasks—from “To Do” through “In Progress” to “Done”, so the entire team can instantly see who’s working on what and where any bottlenecks might be forming. To complement this task-level visibility, we’ve also used a Gantt chart to visualise our overall timeline and key milestones across the full project.

By combining these tools, we have ensured that both day-to-day execution and long-term planning remain synchronised and to quickly grasp our progress and be ready for the upcoming deliverables at a glance.

3.1 Kanban Board (partial)

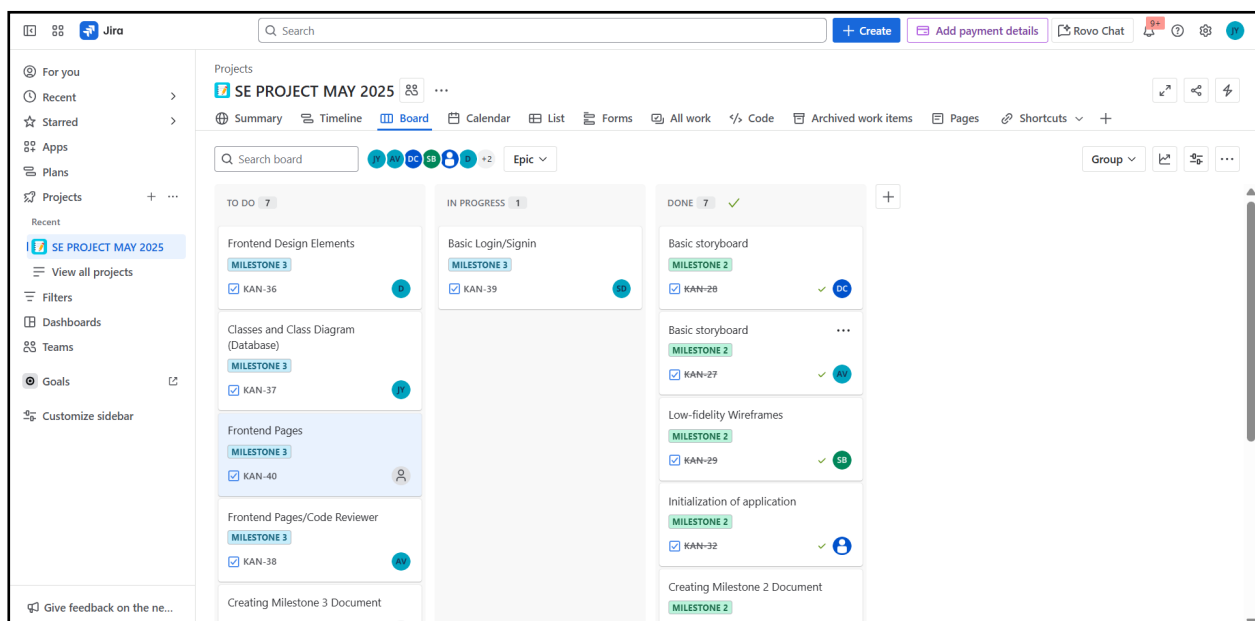


Image 1 shows partial kanban board from Jira

3.2 Gantt Chart

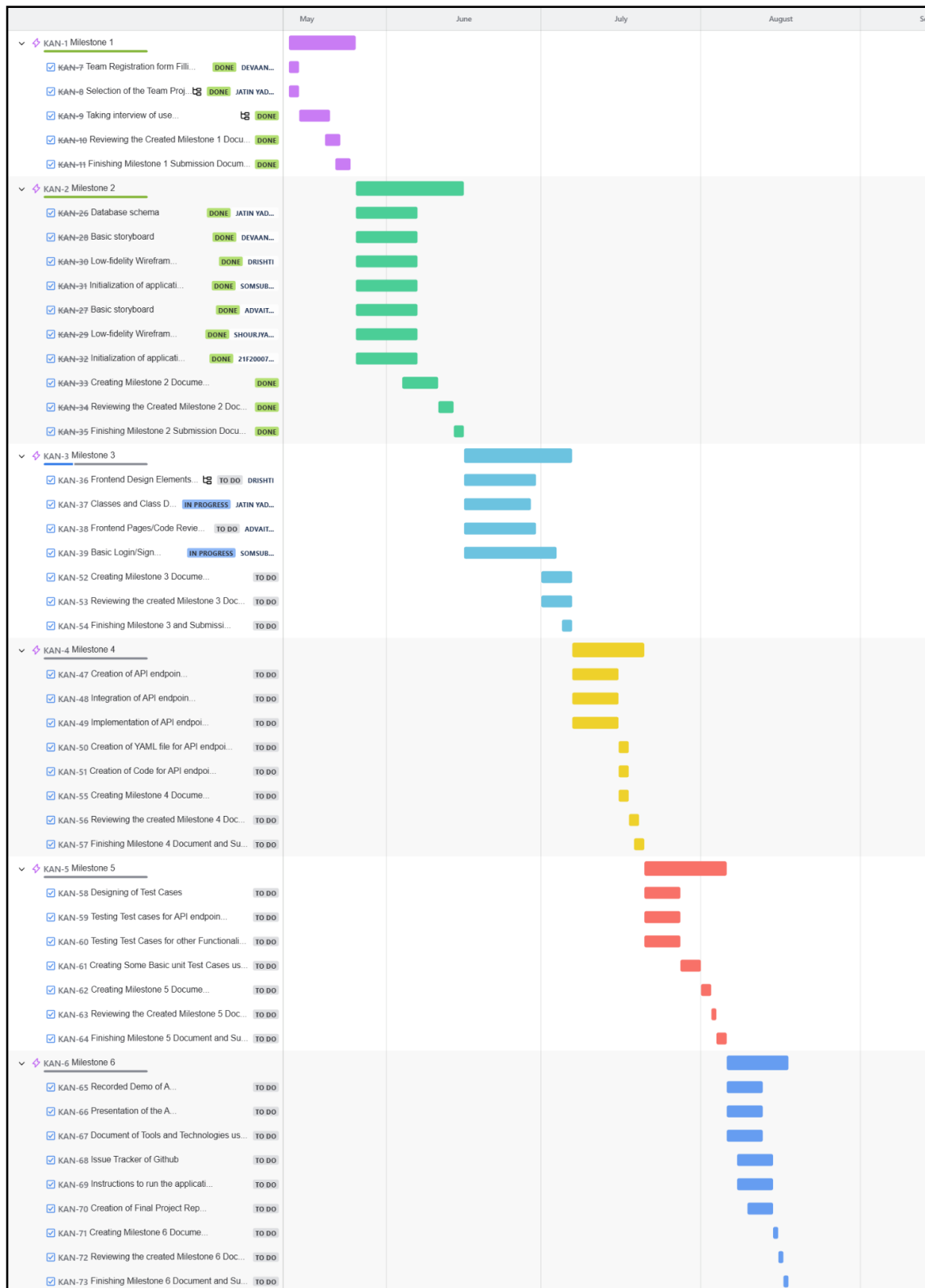


Image 2 shows Gantt chart used to visualise the project timeline

4. Class Diagram

A class diagram is a static UML model that depicts a system's classes, their attributes, and the relationships among them. We have used it to clearly define our domain entities and communicate the structure and associations for our *Saathi* app.

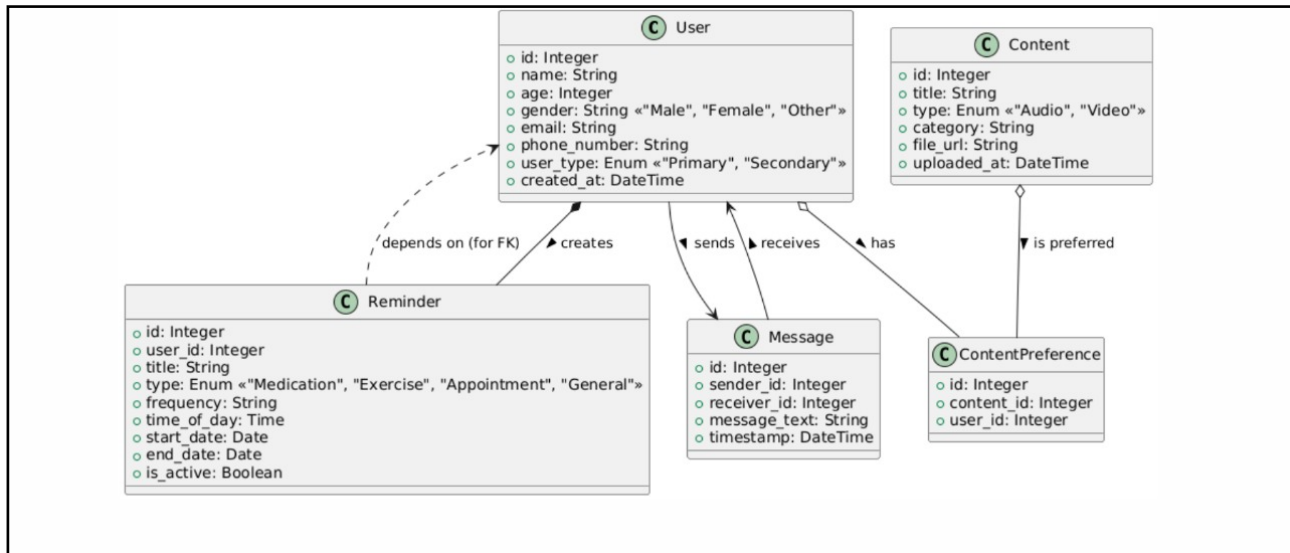


Image 3 shows the class diagram for the app.

I. User

Captures each user's profile with attributes like id, name, age, gender, email, phone_number, user_type (Primary/Secondary), and created_at; serves as the anchor for reminders, messages, and content preferences.

II. Reminder

Represents user-specific notifications (e.g., Medication, Exercise, Appointment, General) with title, frequency, time_of_day, start_date, end_date, and an is_active flag, linked back to its creator via user_id.

III. Message

Models the communication between users by storing sender_id, receiver_id, message_text, and a timestamp, enabling one-to-many "sends" and "receives" relationships with the User class.

IV. Content

Defines multimedia assets (Audio or Video) with id, title, type, category, file_url, and uploaded_at, representing the pool of media we can deliver to users.

V. ContentPreference & Relationships

Acts as a join table (user_id ↔ content_id) to record which content items each user prefers, establishing a many-to-many link between Users and Content and ensuring flexible, personalized media recommendations.

5. Components' Design

Primary User (Senior Citizens) Components

I. Dashboard

- Quick-link cards to all main sections: Entertainment, Health Management, Messaging, Mythology, and AI Voice Assistant
- Accessibility controls (e.g. font-size slider) persistently available

II. Entertainment Page

- List of available simple games (“Memory Match,” “Word Puzzle,” etc.)
- One-click “Play” interface with large tap targets
- Font-size adjustment for all in-game text

III. Health Management Page

- Medication Reminders: Configure daily/weekly schedules with clear labels and toggle switches
- Prescription Upload: Snap or select prescription images, view/manage them in a scrollable list
- Appointment Booking: Simple date-picker flow to schedule doctor visits, with a list of upcoming appointments
- Search Bar: Filter reminders or prescriptions by keyword directly on the page

IV. Messaging Page

- Large-text chat bubbles for sending and receiving messages
- Persistent notification badge or pop-up toast for new incoming messages

V. Mythology Page

- Scrollable library of mythology texts in large, high-contrast font
- Embedded audiobook player with play/pause and skip controls
- Font-size adjustment control for text content

VI. AI Voice Assistant Page

- Microphone-powered interface: “Tap to Speak” button prominently displayed
- Spoken prompts read aloud responses and read back reminders or messages
- Visual transcript area with adjustable font size

Secondary User (Caretaker/ Family Members) Components

I. Dashboard

- Summary cards showing each linked senior's next medication, last acknowledgement, and any missed alerts
- Quick actions to Edit Reminder, Upload Prescription, or Book Appointments.

II. Health Data & Reminders

- Detailed view of a senior's medication schedule with edit/delete controls
- Prescription management on behalf of the senior: upload, label, and archive

III. Appointment Management

Shared appointment calendar to book or modify medical visits for seniors

Notifications for upcoming or overdue appointments

IV. Messaging & Notifications

- View the senior's incoming/outgoing messages in a consolidated feed
- Option to resend reminders or escalate via call/SMS if needed

V. Content & Settings

- Assign or recommend AI-narrated audiobooks and mythology readings to seniors
- Adjust seniors' accessibility settings (font size, voice prompt speed) remotely

6. Frontend

1. Login Page

app

- AI assistant
- Book Content
- Book Library
- Entertainment
- Family Dashboard
- Health
- Home
- Messages app
- Mythology Homepage

Toggle Theme

Dark Mode

Deploy

Welcome to Saathi

Select Option

☒ Login ☐ Sign Up

Login with OTP

Enter your Email ID

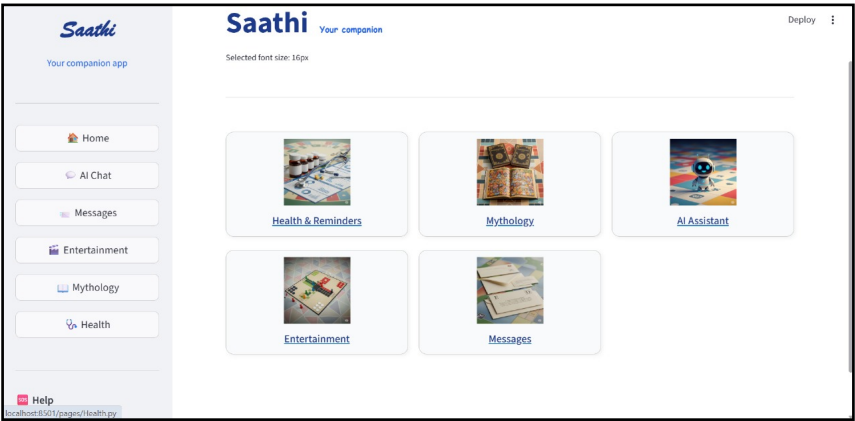
you@example.com

Who are you?

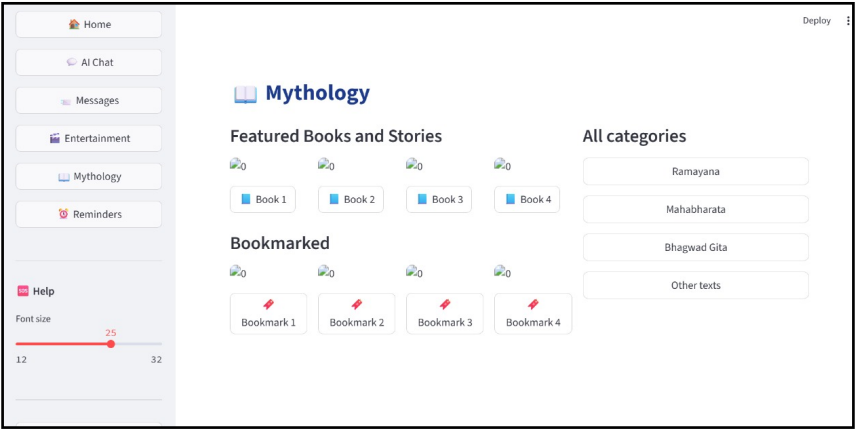
Senior Citizen

Send OTP / Login

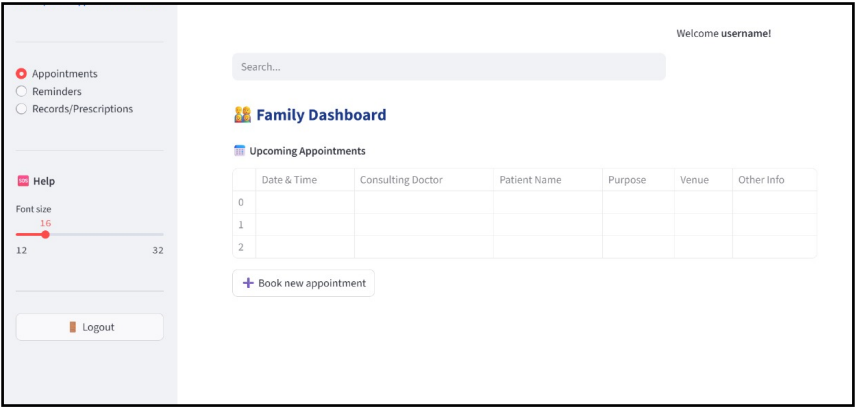
2. Senior Citizen Dashboard



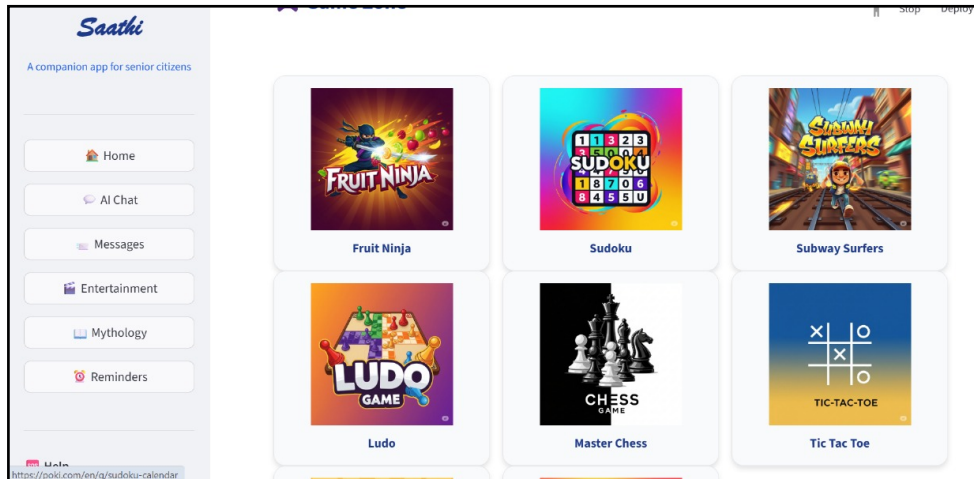
3. Mythology Page



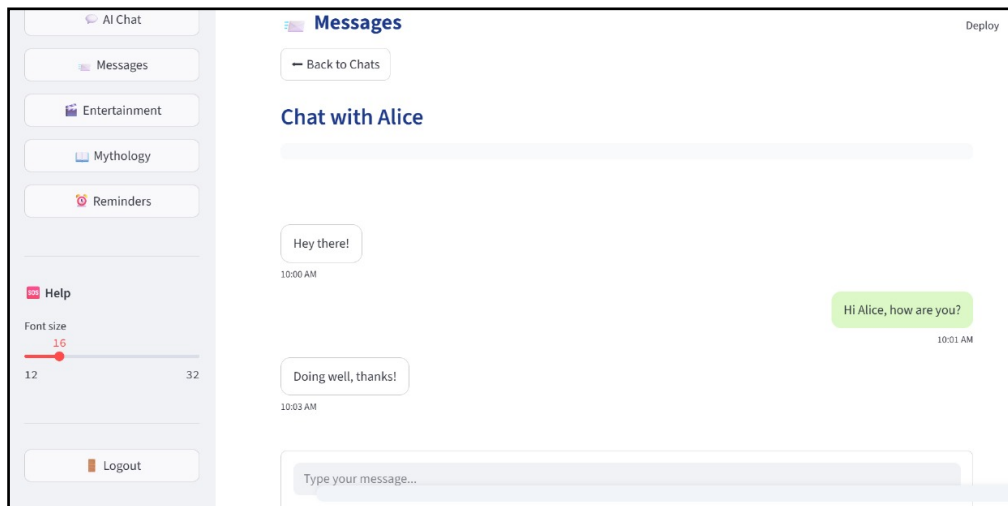
4. Family Dashboard Page



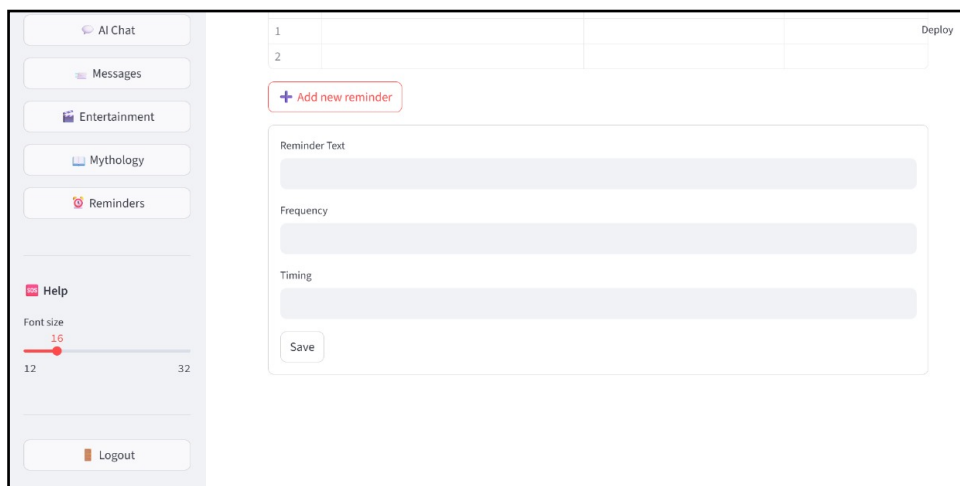
5. Games Page



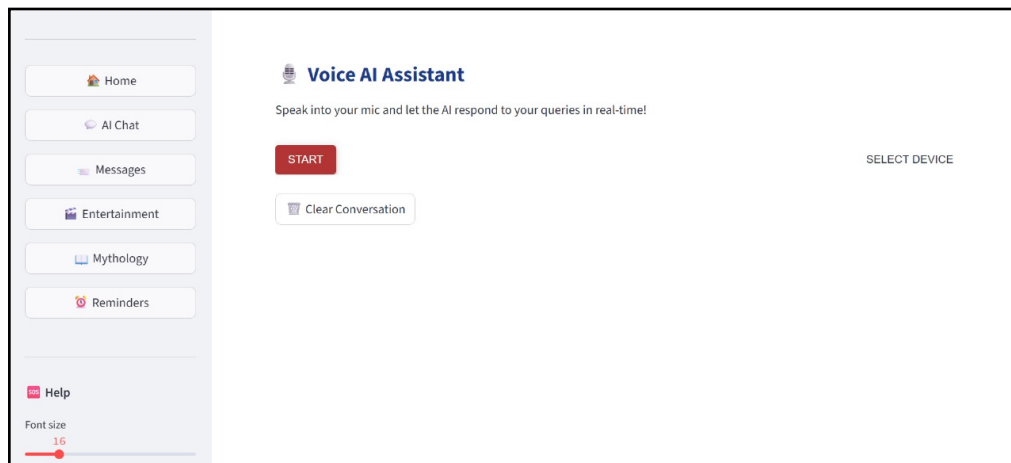
6. Messaging Page



7. Reminder Page



8. AI Voice Assistant



9. Health Dashboard

